

Concomitant assessment of PD-1 and CD56 expression identifies subsets of resting cord blood Vδ2 T cells with disparate cytotoxic potential

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ABSTRACT

Vγ9Vδ2 T lymphocytes are programmed for broad antimicrobial responses with rapid production of Th1 cytokines even before birth, and thus thought to play key roles against pathogens in infants. The process regulating Vδ2 cell acquisition of cytotoxic potential shortly after birth remains understudied. We observed that perforin production in cord blood Vδ2 cells correlates with phenotypes defined by the concomitant assessment of PD-1 and CD56. Bulk RNA sequencing of sorted Vδ2 cell fractions indicated that transcripts related to cytotoxic activity and NK function are enriched in the subset with the highest proportion of perforin⁺ cells. Among differentially expressed transcripts, IRF8, previously linked to CD8 T cell effector differentiation and NK maturation, has the potential to mediate Vδ2 cell differentiation towards cytotoxic effectors. Our current and past results support the hypothesis that distinct mechanisms regulate Vδ2 cell cytotoxic function before and after birth, possibly linked to different levels of microbial exposure.

1. Introduction

Vγ9Vδ2 T (Vδ2) cells survey cholesterol biosynthesis by sensing metabolic intermediates in an innate-like, MHC-unrestricted fashion

[1,2]. The cholesterol metabolites with stimulatory activity, overall defined as phosphoantigens (PAg) due to their pyrophosphate moiety, include high potency microbial molecules, in particular E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HMBPP), and low-potency

Abbreviations: Vδ2 T cells, Vgamma9Vdelta2; TCR, T cell receptor; PD-1, Programmed cell death protein 1; BCG, *Bacille Calmette-Guérin*; PAg, phosphoantigens; IPP, Isopentenyl-pyrophosphate; CB, cord blood; CBMC, cord blood mononuclear cells; PBMC, peripheral blood mononuclear cells; ABP, aminobisphosphonate; ZOL, zoledronic acid monohydrate; IL-2, interleukin 2; HMBPP, E)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate; BTN3A1, Butyrophilin 3A1; IRF8, Interferon Regulatory Factor 8; SD, Standard deviation; PCA, Principal component analysis; FBS, Fetal bovine serum; CPM, Counts per million; DE, Differential expression; FGS, Functional gene set; GSEA, Gene set enrichment analysis; IPA, Ingenuity Pathway Analysis.

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compounds produced by host cells, such as isopentenyl pyrophosphate (IPP) [3–6]. Phosphoantigens are sensed via an inside out mechanism not yet completely elucidated. They bind to an intracellular domain of the membrane molecule Butyrophilin 3A1 (BTN3A1), expressed by a broad range of target/accessory cells [7–11]. PAg binding causes changes in BTN3A1 conformation and membrane mobility/distribution, ultimately resulting in V δ 2 cell activation [7,12–14]. While several details about this process are still unknown, it appears that other butyrophilins interacting with BTN3A1 are also needed for V δ 2 cell activation [15,16].

V δ 2 cells display potent cytotoxic function against diverse infected or malignant cells and employ various strategies to kill their targets [17,18]. Killing of target cells often entails the release of granules containing cytotoxic mediators such as perforin, granzymes and granulysin [19–23]. Potential target cells can be rendered (more) susceptible to V δ 2-mediated killing by inhibition of an enzyme in the mevalonate pathway, the farnesyl pyrophosphate (FPP) synthase, as this results in intracellular accumulation of IPP [24–29]. FPP is effectively inhibited by treatment with aminobisphosphonates (ABPs), drugs routinely employed for cancer chemotherapy and activators of V δ 2 cells [26,30,31]. ABPs such as Zoledronic acid (ZOL) are more effective than PAgS for stimulation of neonatal V δ 2 cells [32–34], thus serve as useful tools to study differentiation and function of cord blood V δ 2 cells *in vitro*.

In neonates born to healthy women, cord blood V δ 2 cells are mostly naïve and display different functional features compared to their adult or infant counterpart [32,33,35–41]. In particular, they have little cytotoxic potential, but quickly acquire it after birth [37–39], in a process likely driven by gut microbial colonization and exposure to environmental microorganisms. The molecular pathway(s) involved in the differentiation of naïve cells into cytotoxic effectors has not been identified, and it is unclear whether the same mechanism(s) regulates cytotoxic potential before and after birth. We therefore assessed the involvement of PD-1, an inhibitory receptor expressed prominently at birth compared to later in life, in the modulation of V δ 2 cell cytotoxic potential at birth.

We previously observed that neonatal V δ 2 cells express the inhibitory receptor PD-1 much longer than their adult counterparts after *in vitro* activation. The engagement of this molecule dampens cord blood V δ 2 cell cytokine production and degranulation, suggesting that it serves as a key regulatory factor for V δ 2 cell function in early life [42]. We also noted that acquisition of adult-like V δ 2 cell cytotoxic potential between birth and 12 months of life coincided with decreasing PD-1 and increasing NKG2A expression [38], consistent with the results of another report, which timed the acquisition of cytotoxic potential to the first ten weeks of life [39]. In the current study, we thus investigate if the expression of PD-1 (or lack thereof) is related to the differentiation of cytotoxic effectors in early life. In our *in vitro* culture system, PD-1 expression, especially in the absence of CD56, displayed a strong negative association with cytotoxic potential, measured as the proportion of perforin⁺ V δ 2 cells. Transcriptomic profiles of sorted V δ 2 subsets revealed that PD-1⁺CD56[−] cells (with the lowest cytotoxic potential based on perforin production) expressed significantly lower levels of several genes related to cytotoxicity and NK function compared to PD-1[−]CD56⁺ cells (the subset with the highest cytotoxic potential), consistent with the flow cytometry results. Overall, PD-1 expression on resting V δ 2 lymphocytes at birth serves as a marker to identify cells less likely to have acquired cytotoxic potential and it becomes a strong predictor of low cytotoxic potential in the absence of CD56 expression.

2. Materials and methods

2.1. Specimen collection and mononuclear cell isolation

Cord blood (CB) samples included in this study were obtained from 37 neonates, of whom 20 females and 14 males; for 3 infants, sex was not recorded. The median gestational age was 39 weeks (range 37–42

weeks).

The specimens were obtained from two sources. In the US, the Department of Obstetrics, Gynecology and Reproductive Services, University of Maryland Medical Center provided de-identified specimens. Women were excluded if they tested positive for HIV, HCV, or HBV, were immunosuppressed, had autoimmune disease, had proteinuric pre-eclampsia, used illegal drugs during pregnancy, had traveled to a malaria-endemic region during pregnancy, or if the newborn was below the 10th percentile weight for gestational age.

In Malawi, CB samples were collected from women with no HIV infection who had been enrolled in a clinical trial at the Ndirande Government Health Center, in Blantyre ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT01443130) Identifier: NCT01443130). All specimens included in this study were obtained from term, uncomplicated pregnancies and women had no evidence of malaria infection during pregnancy.

Adult peripheral blood specimens were purchased from the New York blood bank or collected from healthy volunteers at the University of Maryland School of Medicine.

Cord blood (15–60 ml) was collected from the umbilical vein soon after delivery of the placenta using a sterile cord blood collection unit (Pall medical, Port Washington, NY). Cord blood or peripheral blood was diluted with phosphate-buffered saline, PBS (Lonza, Walkersville, MD) and layered over Lymphocyte Separation Medium (LSM, Corning, NY) density gradient to isolate cord blood or peripheral blood mononuclear cells (CBMC and PBMC, respectively). CBMC or PBMC were frozen in 90 % fetal bovine serum (FBS, Gibco, Thermo Fisher Scientific, Waltham, MA), 10 % DMSO (Millipore Sigma, St. Burlington, MA) freezing medium. All specimens were stored in a liquid nitrogen locator before use.

In terms of ethical approvals, the de-identified cord blood specimens in the US were collected in the context of a protocol that was exempt from review by the University of Maryland School of Medicine Institutional Review Board (HP-00056158). The clinical trial in Malawi and related study procedures were approved by the Institutional Review Board of the University of Maryland School of Medicine and by the College of Medicine Research Ethics Committee of the University of Malawi College of Medicine in Blantyre, Malawi. Collection of peripheral blood specimens from adult healthy volunteers at the University of Maryland School of Medicine was approved by the University Institutional Review Board.

2.2. Cell culture

In vitro stimulation of V δ 2 cells was performed as previously described [38]. Briefly, after thawing, CBMC or PBMC pellets were incubated with DNase (Stem Cell Technologies, Vancouver, Canada) then resuspended in RPMI 1640 supplemented with 10 % FBS (Gibco, Thermo Fisher Scientific), 2 mM L-glutamine, and 10 μ g/ml gentamicin (Gibco, Thermo Fisher Scientific) for counting. Cell number was determined by staining with anti-CD45 for six minutes at room temperature and counting the events in the mononuclear cell gate on a Guava easy-Cyte 5HT, (Luminex, Austin, TX). After adjusting the cell concentration to 1.5×10^6 cells/ml, cultures were treated with zoledronic acid monohydrate (ZOL, Millipore Sigma) at 0.5 μ M, or BCG at a pre-optimized multiplicity of infection (MOI) (Pasteur strain, provided by Dr. Manganeli), in the presence of 100 IU/ml of human recombinant interleukin 2 (IL-2) (Tecin, NIH reagent program, NIH, Bethesda, MD). IL-2 alone (100 IU/ml) was used as a negative control treatment. After a washing step, BCG was added to each CBMC and PBMC well at the MOI of 1. Cells were incubated for 14 days at 37 °C with 5 % CO₂ and fresh cytokine was added every 3 days; one volume of fresh medium was added on days 7 and 10. On day 14 a fraction of the CBMC or PBMC was used to determine V δ 2 cell frequency and phenotype; on day 16 an aliquot was used to assess function (degranulation and intracellular cytokine production); on days 10 or 14 cells were sorted for RNA extraction.

2.3. Flow cytometry

Ex vivo or cultured CBMC were resuspended in PBS, stained with a fixable viability dye (BD Horizon™ 780, BD Biosciences, Franklin Lakes, NJ or Zombie Aqua, Biolegend, San Diego, CA) at room temperature for 15 min, and washed twice with PBS-10 % FBS. Afterward, cells were stained at 4 °C for 15 min with the directly conjugated monoclonal antibodies listed below. After washing with PBS, cells were resuspended in PBS-0.4 % paraformaldehyde. $(3-8) \times 10^5$ lymphocytes (gated based on forward and side scatter profiles) were collected for each specimen on an Aria II or LSR II (BD Biosciences). The acquisition files were analyzed with FlowJo software (FlowJo LLC, BD, Ashland, OR). Vδ2⁺ T cells were manually gated among CD3⁺ cells after excluding doublets and dead cells from the lymphocyte gate.

The monoclonal antibodies used for polychromatic cell surface staining are listed below. Anti-Vδ2 (clone B6), anti-PD-1 (clone EH12.2H7), anti-CD69 (clone FN50), anti-CD3 (clone OKT3, UCHT1), anti-CD16 (clone 3G8), anti-CD56 (clone 5.1H11), were purchased from Biolegend (San Diego, CA). Anti-CD3 (clone BW264/56), anti-PD-1 (clone PD-1.3.1.3), anti-CD16 (clone REA423), anti-CD45RA (clone T6D11), anti-CD45RO (clone UCHL1), anti-CD56 (clone REA196), anti-NKG2A (clone REA110), anti-NKG2D (clone BAT221), were obtained from Miltenyi Biotec (Bergishch Gladbach, Germany). Anti-CD25 (clone CD25-4E3), and anti-HLA-DR (clone L243) were purchased from Thermo Fisher Scientific. Anti-CD4 (clone SK3) and anti-CD8 (clone RPA-T8) were obtained from BD Biosciences.

2.4. Production of cytotoxic mediators and IRF8 expression

To evaluate Vδ2 cell cytotoxic potential *ex vivo* and after stimulation, we monitored the production of cytotoxic mediators as previously described [38,42]. Briefly, CBMC or PBMC (*ex vivo* or 16 days after stimulation) were counted, stained for viability and surface markers (as described above), then permeabilized by incubation with fixation/permeabilization solution (BD Biosciences) at 4 °C for 20 min. The cells were then incubated for 40 min at room temperature with anti-Perforin (clone DG9, Thermo Fisher Scientific), anti-Granulysin (clone DH2, Biolegend), and anti-Granzyme A (clone CB9, Biolegend) or anti-Granzyme B (clone GB11, Biolegend) diluted at preoptimized concentrations in Perm/wash buffer (BD Biosciences). After washing, the cells were resuspended in a 0.4 % PFA solution for acquisition. At least 8×10^5 events *ex vivo* and 3×10^5 after expansion were collected in the lymphocyte gate for each sample on a FACS Aria II (BD Biosciences).

A small subset of specimens was stained to assess IRF8 expression relative to PD-1, CD56 and perforin. In these experiments, 7 and 10 days after ZOL stimulation the CBMC were counted, stained for viability and surface markers (as described above), then permeabilized by incubation with eBioscience FoxP3 fixation/permeabilization solution (Invitrogen) at 4 °C for 40 min. The cells were then washed with permeabilization solution and incubated for 40 min at room temperature with anti-Perforin (clone DG9, Thermo Fisher Scientific) and anti-IRF8 (clone U31-644, BD Biosciences). At least 1.5 million viable lymphocytes were acquired on a 5 laser Aurora analyzer (Cytek Biosciences, Fremont, CA).

2.5. Cytokine production and granule mobilization assay

Sixteen days after stimulation, CBMC or PBMC were resuspended at 2×10^6 cells/ml in fresh complete medium and re-stimulated in 96-well plates pre-coated with anti-γδ TCR (clone B1.1, Thermo Fisher Scientific). The plates were coated overnight at 4 °C with anti-γδ TCR (diluted 1:100 in PBS, 50 μl/well). Cells were plated in triplicate (100 μl/well) in the presence of anti-CD107a AlexaFluor488 (3 μl/well, clone H4A3, Biolegend), GolgiPlug (brefeldin A, 1 μg/ml, BD Biosciences), and GolgiStop (monensin, 1 μg/ml, BD Biosciences). After a 6-hour incubation, the cells were collected, washed once with cold PBS, and stained with a viability dye, followed by surface staining as described above. Cells were

then permeabilized by incubating for 20 min at 4 °C with fixation/permeabilization solution (BD Biosciences) and washed twice with Perm/wash buffer (BD Biosciences). Anti-human TNFα APC (clone Mab11, BD Biosciences) and anti-IFNγ PE-Vio770 (clone 45-15, Miltenyi Biotec) were added for 40 min at room temperature. At least 3×10^5 lymphocytes were collected for each sample on a FACS LSR II (BD Biosciences).

2.6. RNA extraction from sorted Vδ2 cell subsets

Ten or fourteen days after stimulation, CBMC and adult PBMC were stained with anti-Vδ2 (clone B6), anti-PD-1 (clone EH12.2H7), anti-CD56 (5.1H11), washed twice and sorted into PD-1⁺CD56⁺ and PD-1⁺CD56⁻ fractions on an FACS Aria II sorter, with >95 % purity. The small aliquot of cells used for quality control was sorted in FBS, the cells for RNA extraction [$(3-12) \times 10^5$ for RT-PCR and $(8-10) \times 10^5$ for bulk RNA-seq], were sorted in Trizol. RNA was extracted using Direct-zol RNA microprep kit (Zymo Research, Irvine, CA). The nucleic acid, eluted in molecular biology grade water, was quantified using a Nano-Drop spectrophotometer (Thermo Fisher Scientific) and immediately converted into cDNA. Leftover RNA was cryopreserved at -80 °C.

2.7. RT-PCR

Total RNA (0.5 μg) was converted into cDNA using the iScript cDNA Synthesis kit (Bio-Rad Laboratories, Hercules, CA) following manufacturer's instruction. cDNA was diluted with one volume of nuclease-free water for amplification of target genes, while further 1/1000 dilutions were applied for amplification of reference genes. Real-time PCR reactions were set up according to SsoAdvanced Universal SYBR Green Supermix protocol (BioRad Laboratories). The amplifications were performed and analyzed using QuantStudio3 (ThermoFisher Scientific) and implementing the standard cycling program: 50 °C for 2 min followed by 95 °C for 10 min; then 40 cycles of denaturation at 95 °C for 15 s and extension at 60 °C for 1 min. Primer sequences for target genes include: IRF8 forward 5'-AGCAAGATTATAATCAGGAAGTGGA-3' and reverse 5'-TCAAAGCACAGCGTAACCTC-3'; zbtb32 forward 5'-GACAGGGCTAAAAAGCCAGAT-3' and reverse 5'-GCGAGTGCTTGTC-CAACATC-3'. Two non-overlapping 18S ribosomal RNA segments were used as reference controls, amplified by the following primer pairs: 18S1 forward 5'-AGAAACGGCTACCACA-3', reverse 5'-CACCA-GACTTGCCCTCCA-3'; 18S2: forward 5'-CAGCCACCCGAGATTGAGCA-3', reverse 5'-TAGTAGCGACGGGCGGTGTG-3'. Primer concentration was 200 nM for target genes, 350 and 300 nM for 18S1 and 18S2 respectively. The efficiency of each primer pair is within the recommended range (90 %–110 %). The variation between runs was normalized using the same tonsil cDNA specimen as internal reference. The relative expression of target genes was analyzed using the comparative method [$2^{-(\Delta\Delta Ct)}$] [43].

2.8. Transcriptomic analysis

Total RNA (1 μg/specimen, eluted in water) samples were shipped to Novogene in dry ice for bulk RNA sequencing. STAR (v. 2.7.8a) was used to align transcriptomic sequences to Genome Reference Consortium Human (GRCh38) Build 38 with GRCh38 104 annotations. Gene expression was then quantified with *featureCounts* (subread v2.0.1). All remaining analyses were performed within the R/Bioconductor environment (v4.0.1) [44–46].

The R/Bioconductor package *edgeR* [47] (v3.34) was used to filter low count genes, normalize by library size, and calculate log2 counts per million (log-CPM) gene expression values. To this end, genes at low expression levels were removed as previously described [48], and the M-value trimmed mean method was used for library size normalization and log-CPM computation [48,49].

Linear modelling and differential expression (DE) analysis were

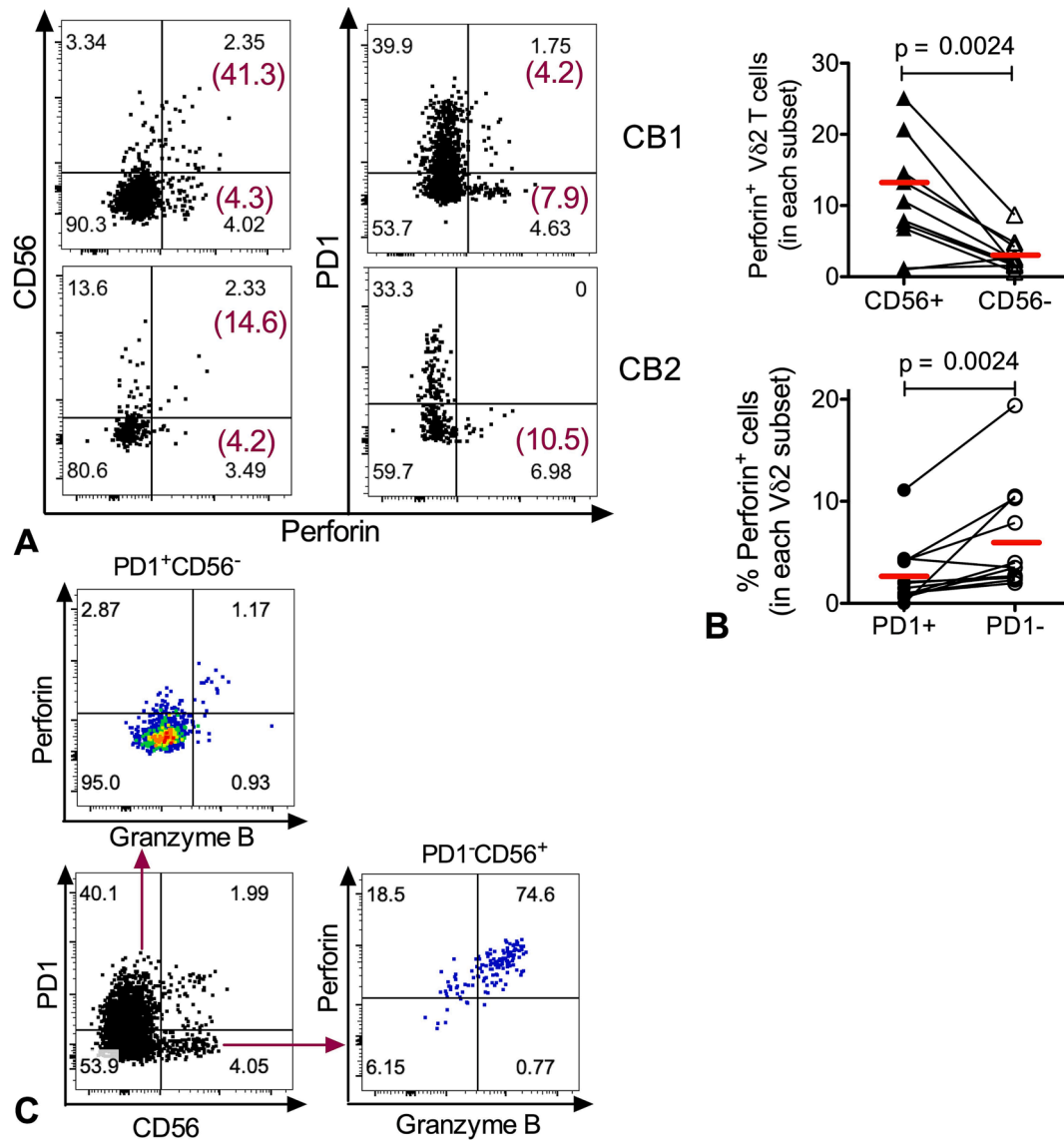


Fig. 1. PD-1⁻ and CD56⁺ cord blood Vδ2 lymphocytes *ex vivo* are enriched in perforin⁺ cells. CBMC were thawed and stained to assess *ex vivo* expression of surface markers and cytotoxic mediators. A) The dot plots display the expression of perforin in relation to CD56 or PD-1 for two representative CBMC specimens. The percentages in parentheses represent the proportion of perforin⁺ cells in the CD56⁺ versus CD56⁻ subset or in the PD1⁺ versus PD1⁻ subsets of Vδ2 cells. B) The plots show the proportion of perforin⁺ cells in paired subsets of Vδ2 cells (CD56⁺ and CD56⁻ on the left, PD1⁺ and PD1⁻ on the right) for 10 donors. C) The dot plot and pseudocolor plots display the *ex vivo* expression of perforin and granzyme B relative to PD-1 and CD56 (in PD-1⁻CD56⁺ and PD-1⁺CD56⁻ Vδ2 cells) for a cord blood specimen with an above-average proportion of perforin⁺ Vδ2 cells.

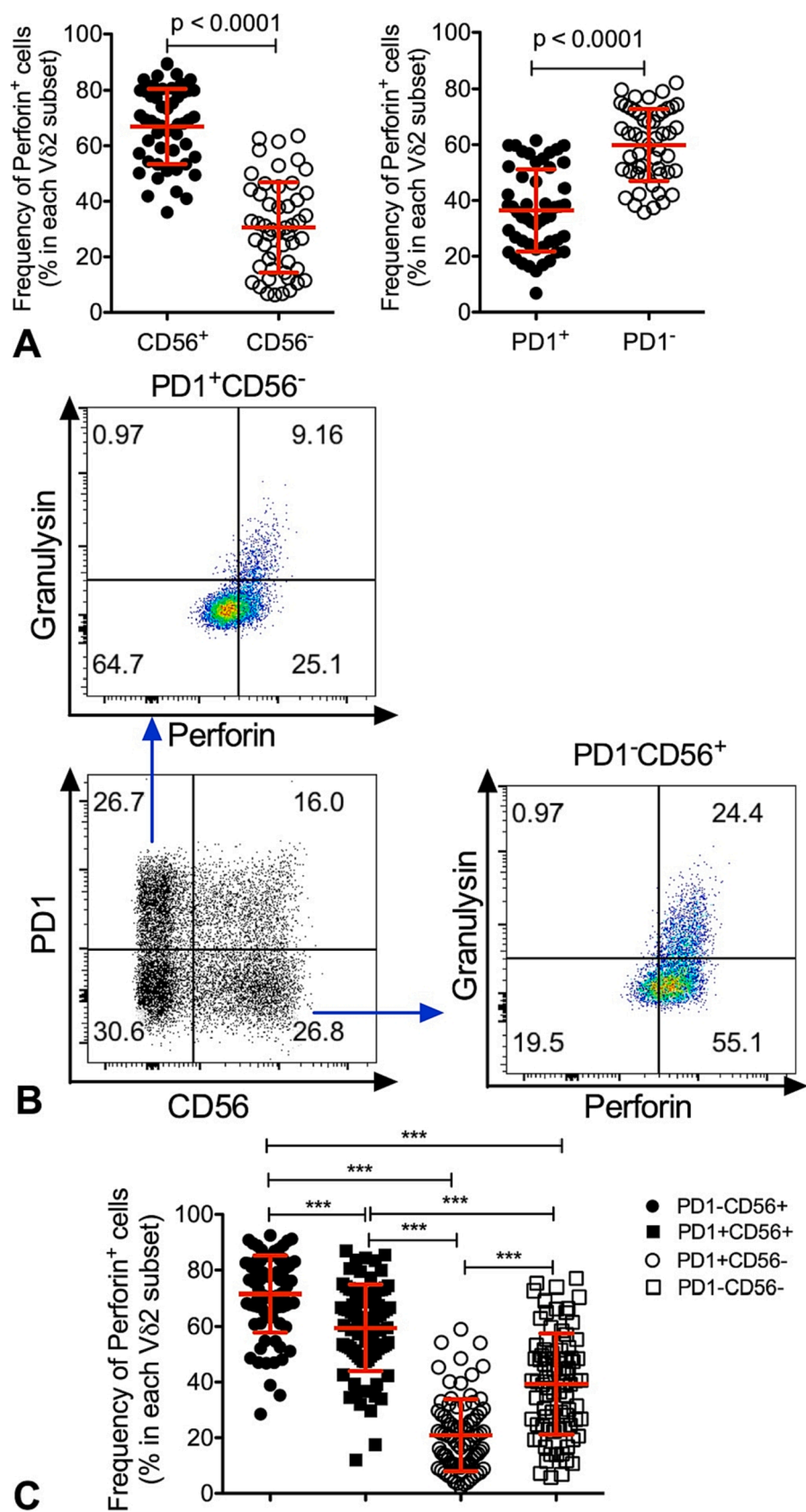
performed with *limma* (v3.48.3) using a paired sample design [50]. In particular, mean-variance relationship and appropriate observation-level weights were estimated on quantile-normalized log-CPM. For each gene, the linear model was fit on expression data and moderated *t*-statistics, moderated F-statistic, and log-odds of differential expression were computed by empirical Bayes moderation of the standard errors towards a global value [50,51].

2.9. Gene set enrichment analysis

Gene set enrichment analysis (GSEA) was performed to identify pathways and biological processes (*i.e.*, Functional Gene Set, FGS) associated with the phenotype of interest, using a two-pronged approach. Systematic identification of enriched FGS was performed using *fgsea* (v7.4.1) [52–54], while the Ingenuity Pathway Analysis (IPA) tool (<http://www.ingenuity.com>) was used to further deepen functional inference derived from DE analysis. Briefly, for GSEA using

fgsea, genes were ranked according to *t*-statistics from the DE analysis, and an adaptive multilevel split Monte Carlo approach with 1000 permutations was adopted to estimate enrichment scores and relative *p*-values [55]. In this analysis, FGS were deemed significant compared to a null background obtained by permutation ($n = 200$), as previously described [56]. Data wrangling and visualizations were performed using *tidyverse* packages [57].

The IPA tool was used to map and generate putative enriched networks based on manually curated knowledge of pathway interactions as derived from the literature. Pathways were ranked by significance scores that measured the probability of genes to be included in the pathway by chance. Specifically, results were ranked by statistical significance as obtained from a hypergeometric test. For all analyses, multiple testing correction was performed using the Benjamini-Hochberg method, and an adjusted *p*-value less than 0.01 was used as a cutoff to select significantly enriched pathways.



(caption on next page)

Fig. 2. Concomitant assessment of PD-1 and CD56 identifies neonatal Vδ2 cell subsets with different cytotoxic potential. CBMC were stimulated with ZOL (5 μM) or BCG (MOI = 1) plus IL-2 (100 U/ml) and cultured for 15 days. On day 16, the cells were stained to assess the presence of cytotoxic mediators in relation to surface marker expression. A) The scatter plots show the proportion of perforin⁺ cells in paired subsets of Vδ2 cells (CD56⁺ and CD56⁻ on the left, PD-1⁺ and PD-1⁻ on the right) with mean, SD and individual values. B) The dot plot and pseudocolor plots display the expression of perforin and granulysin in cord blood PD-1⁻CD56⁺ and PD-1⁺CD56⁻ Vδ2 cells for a representative specimen stimulated with ZOL. C) The scatter plot depicts the proportion of perforin⁺ Vδ2 cells across the four subsets identified by CD56 and PD1, showing mean, standard deviation and individual values for each subset.

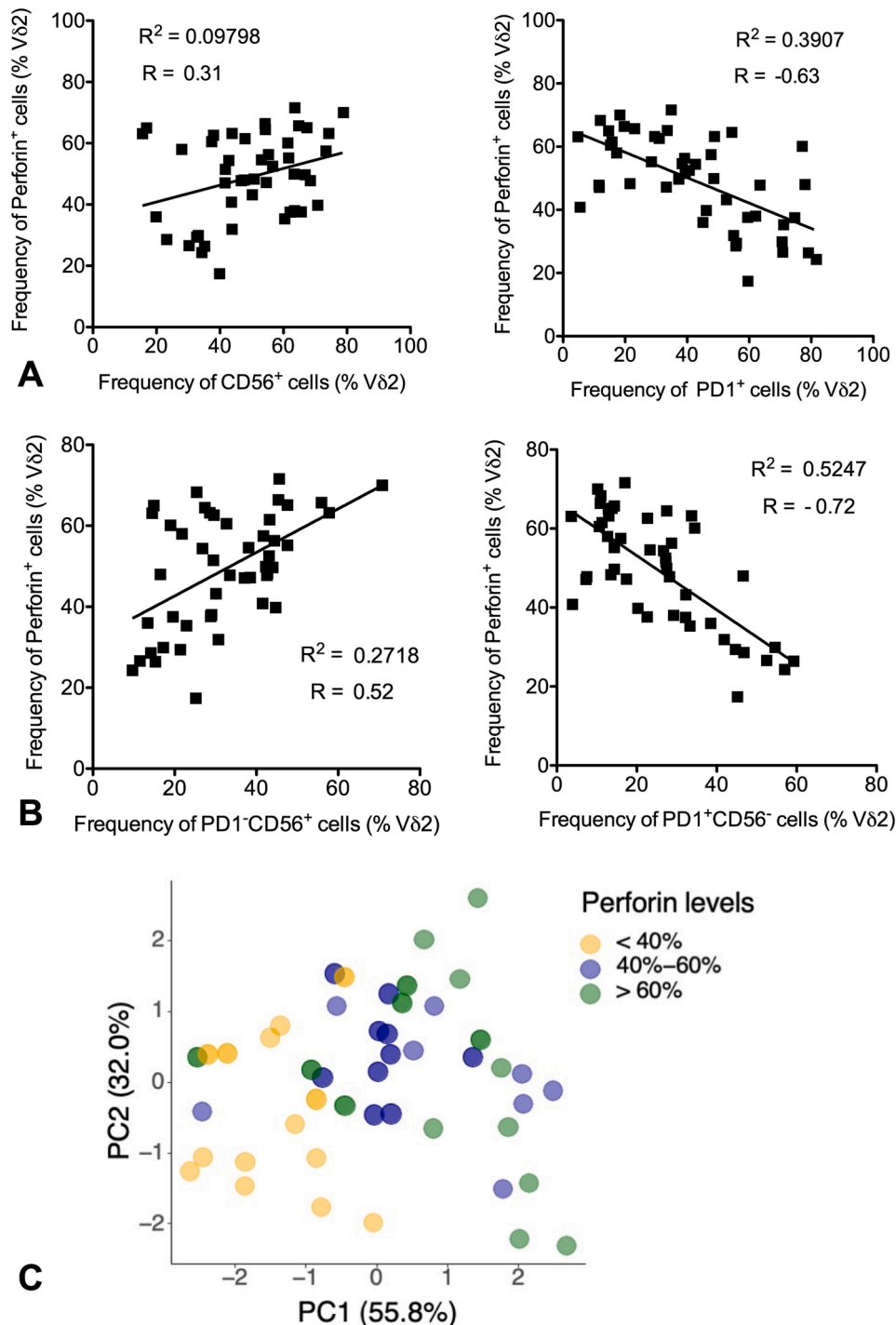


Fig. 3. Perforin expression correlates with the expression of specific surface markers. On day 16 after stimulation, CBMC were stained to assess the presence of cytotoxic mediators relative to surface marker expression; simple linear regression was employed to test for correlations. A) The linear regression plots display the relationship between the proportion of perforin⁺ and the proportion of total CD56⁺ (left) or PD-1⁺ (right) cells in cord blood Vδ2 lymphocytes. B) The linear regression plots display the correlation between the proportion of perforin⁺ and the proportion of PD-1⁻CD56⁺ (left) or PD-1⁺CD56⁻ (right) cells in cord blood Vδ2 lymphocytes. The PCA plots show the distribution of CBMC based on the distribution of cells across the 4 subsets identified by CD56 and PD-1.

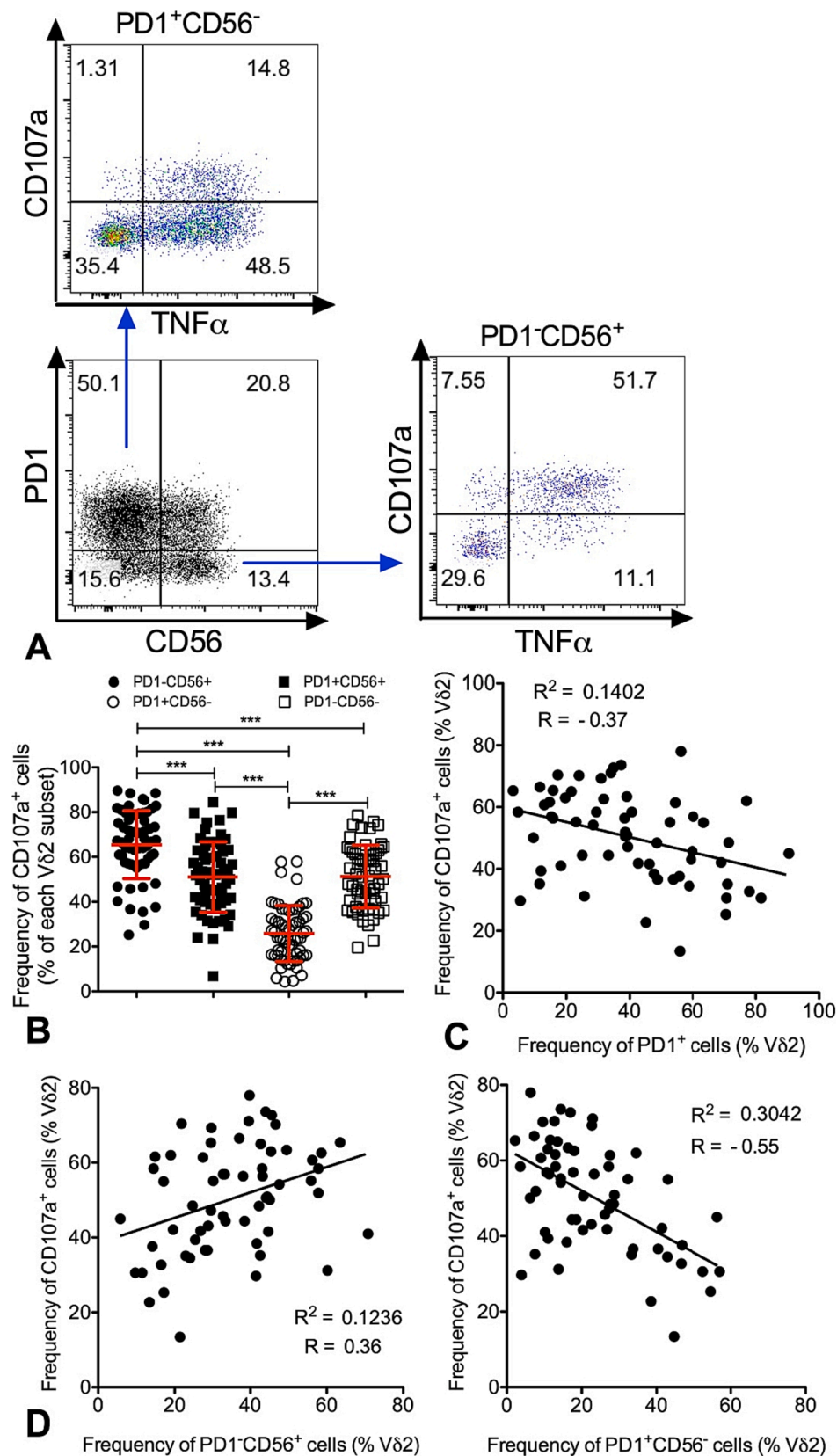


Fig. 4. CD56 and PD-1 identify Vδ2 cell subsets with different membrane CD107a expression. Sixteen days after stimulation, CBMC were restimulated for six hours with plastic-immobilized anti $\gamma\delta$ TCR (clone B1.1) in the presence of anti-CD107a. A) The dot plot and pseudocolor plots depict the expression of CD107a and TNF α in PD-1⁻CD56⁺ and PD-1⁺CD56⁻ Vδ2 cells for a representative donor. B) The scatter plot depicts the proportion of CD107a⁺ Vδ2 cells across the four subsets identified by CD56 and PD-1, showing mean, standard deviation and individual values for each subset. C) The linear regression plot displays the relationship between the proportion of CD107a⁺ and the proportion of total PD-1⁺ cells in cord blood Vδ2 lymphocytes. D) The linear regression plots display the correlation between the proportion of CD107a⁺ and the proportion of PD-1⁺CD56⁺ (left) or PD-1⁺CD56⁻ (right) cells in cord blood Vδ2 lymphocytes.

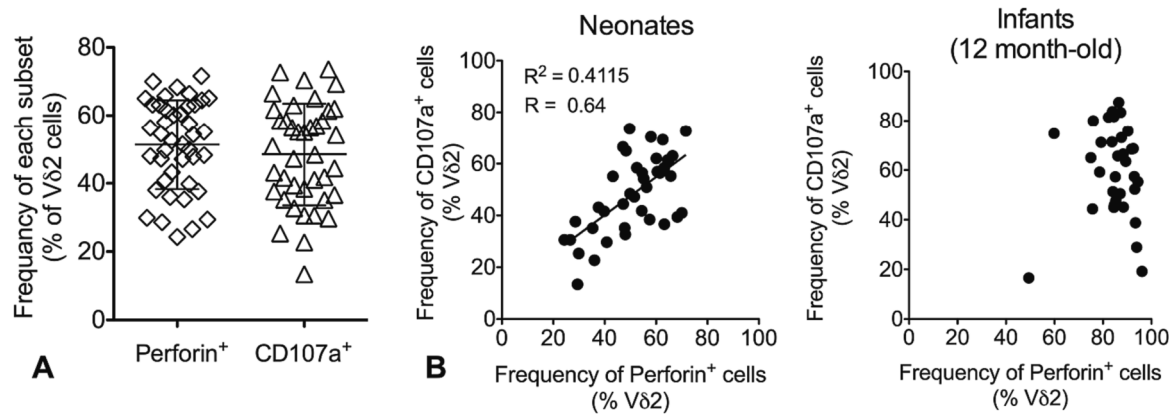


Fig. 5. The proportion of CD107a⁺ cells correlates with the proportion of perforin⁺ cells in neonatal but not infant Vδ2 cells. Sixteen days after stimulation, CBMC or infant PBMC were assessed for expression of cytotoxic mediators and for granule mobilization following TCR-mediated restimulation. A) The scatter plot compares the proportion of perforin⁺ and CD107a⁺ cord blood Vδ2 cells, showing mean, standard deviation and individual values for each variable. B) The linear regression plots display the correlation between proportion of perforin⁺ and CD107a⁺ Vδ2 cells derived from cord blood (left) or infant peripheral blood (right).

2.10. Statistical analysis

Statistical analyses were performed using the software GraphPad Prism V5.0f (GraphPad Software, La Jolla, CA) and R (v 4.1.1, with the packages FactoMineR and Factoextra) [46,58,59]. For each variable, D'Agostino & Pearson omnibus normality test was employed to assess whether values were normally distributed. Differences between means (for normally distributed variables) or medians (for variables displaying a non-Gaussian distribution) were evaluated, respectively, using a student's *t*-test or Mann-Whitney test for unpaired groups, and paired *t*-test or Wilcoxon test for paired groups. When we could not test for normality due to small sample size, we used non-parametric tests.

We used compositional analysis to evaluate the relationship between perforin expression and the 4 subsets generated by PD-1 and CD56 in a flow cytometry plot (Q1: PD-1⁺CD56⁺, Q2: PD-1⁺CD56⁺; Q3: PD-1⁺CD56⁺, Q4: PD-1⁺CD56⁺). We created a principal component analysis plot in R based on eigen values (obtained reducing Q1-Q4 frequencies) and stratified perforin in three levels (<40 %, 40 %–60 % and >60 % frequency). Linear regression analyses, both single and multi-variable models, were used to assess the relationship between perforin levels (% positive cells) and proportion of PD-1⁺CD56⁺, PD-1⁺CD56⁺ and CD107a⁺ cells. Coefficients, estimates and 95 % confidence intervals were calculated, and *p* < 0.05 was considered significant.

3. Results

3.1. Cord blood Vδ2 cells expressing PD-1 are less likely to store perforin than those not expressing PD-1

Expression of some surface markers is known to be associated with better cytotoxic potential in adults and infant Vδ2 cells [19,38,60]. In cord blood, perforin⁺ cells were enriched in the CD56⁺ subset of Vδ2 lymphocytes, as expected based on previous reports [19], but also in the PD-1⁺ subset (Fig. 1 and [38]). We thus sought to further explore the relationship between PD-1 and perforin expression. In the few specimens with numbers of perforin⁺ cells high enough to enable a breakdown of perforin production by CD56 and PD1 expression, we observed that the PD-1⁺CD56⁺ subset displayed the highest proportion of perforin⁺ cells, while the PD-1⁺CD56⁺ subset contained the lowest (93 % and 4 %, respectively in Fig. 1 C). However, due to the small numbers of perforin⁺ cord blood Vδ2 cells *ex vivo*, we could not perform this type of analysis for most specimens. We therefore resolved to expand Vδ2 cells with either zoledronic acid (ZOL) or *Bacille Calmette-Guerin* (BCG), as described previously [38] to obtain enough cells. For all subsequent analyses, we pooled the results of specimens expanded with ZOL or BCG,

because the two stimuli, while yielding substantially different proportions of PD-1⁺ cells, shared similar trends for perforin expression in relation to PD-1 [38]. The trend was not observed for cells that had undergone IL-2-driven homeostatic proliferation nor for 12-month-old infant cells (data not shown).

After expansion, perforin expression patterns were similar to those observed *ex vivo*, with perforin⁺ cells more abundant in the PD-1⁺ and CD56⁺ subset of Vδ2 lymphocytes (Fig. 2 A). Analysis of perforin storage based on concomitant assessment of CD56 and PD-1 expression in a large number of specimens also confirmed the *ex vivo* observation that the PD-1⁺CD56⁺ subset contains the highest proportion of perforin⁺ cells (71.6 %), while the PD-1⁺CD56⁺ subset has the lowest (20.9 %, Fig. 2 B-C). The double positive and double negative subsets contained intermediate frequencies of perforin⁺ cells. Expression of granulysin followed a similar trend, though frequencies of positive cells were overall lower (20 % in the PD-1⁺CD56⁺ subset versus 7.1 % in the PD-1⁺CD56⁺ subset, Fig. S1). Importantly, cytokine production after short-course TCR-mediated restimulation did not follow the same trend: the proportion of TNFα⁺ cells was similar across subsets and highest in the PD-1 single positive subset, while the proportion of IFNγ⁺ cells was highest in the two CD56⁺ subsets regardless of PD-1 expression (Fig. S1).

For a subset of specimens, we analyzed expression of CD4 and CD8 relative to CD56 and PD-1 after expansion with ZOL (Fig. S2). The PD-1⁺CD56⁺ subset was mostly CD8⁺ or double negative, with few or no CD4⁺ cells. On the opposite, PD-1⁺CD56⁺ cells were mostly CD4⁺ or double negative, with relatively few CD8⁺ cells.

3.2. Expression of PD-1 and CD56 correlates with the frequency of perforin⁺ Vδ2 cells

We assessed whether a relationship exists between the proportion of Vδ2 cells expressing PD-1 or CD56 and the proportion of perforin⁺ cells using a linear regression model. The percentage of perforin⁺ Vδ2 cells was weakly correlated with the percentage of CD56⁺ cells (*R* = 0.31, *p* = 0.034) and displayed a moderate negative correlation with the percentage of PD-1⁺ cells (*R* = −0.63, *p* < 0.0001, Fig. 3 A). When we assessed the correlation of perforin with each of the 4 subsets identified by CD56 and PD-1, the percentage of perforin⁺ cells showed a strong negative linear correlation with the percentage of PD-1⁺CD56⁺ cells and a moderate positive linear correlation with the percentage of PD-1⁺CD56⁺ cells (*R* = −0.72 and 0.52 respectively, with *p* < 0.0001, and *p* = 0.0002 respectively, Fig. 3 B). Principal component analysis (PCA) was performed to test whether specimens with different proportions of perforin⁺ Vδ2 cells cluster separately based on the composition of the four subsets identified by CD56 and PD1. We stratified the percentage of

Table 1
Multivariable regression models.

	Outcome	Variables	Estimate	t value	p-value	Multiple R ²
Model 1	Perforin	PD-1 ⁺ CD56 ⁺	0.2782	2.430	0.0201	0.4925
Model 2	Perforin	CD107a	0.4701	4.328	0.0001	0.5491
Model 3	CD107a	PD-1 ⁺ CD56 ⁺	-0.4039	-3.360	0.0018	0.4126
Model 4	CD107a	Perforin	0.3437	2.969	0.0052	0.4126
Model 5	PD-1 ⁺ CD56 ⁺	CD107a	0.7150	4.328	0.0001	0.4412
Model 6	PD-1 ⁺ CD56 ⁺	Perforin	-0.2394	-1.402	0.1693	0.2369
Model 7	PD-1 ⁺ CD56 ⁺	CD107a	0.5597	2.969	0.0052	0.2369
Model 8	PD-1 ⁺ CD56 ⁺	Perforin	0.4946	2.430	0.0201	0.2369
Model 9	PD-1 ⁺ CD56 ⁺	CD107a	0.0450	0.254	0.8012	0.4699
Model 10	PD-1 ⁺ CD56 ⁺	Perforin	-0.5788	-3.360	0.0018	0.4699
Model 11	PD-1 ⁺ CD56 ⁺	CD107a	-0.2107	-1.402	0.1692	0.4699

perforin⁺ Vδ2 cells in 3 levels: <40 % (low), 40–60 % (medium), >60 % (high). The first component PC1, explained 55.8 % of variance while the second component PC2 explained 32 % of variance in the dataset. The main loading for PC1 was contributed by the percentage of PD1⁺CD56⁺ cells, and the main loading for PC2 was contributed by the percentage of PD1⁺CD56⁺ cells. Most of the specimens in the low perforin stratum (<40 %) clustered separately from specimens in the other two strata (Fig. 3 C). Other NK receptors, such as NKG2A and CD16, identified subsets enriched in perforin⁺ cells both *ex vivo* and after expansion, but there was no significant correlation between the proportion of perforin⁺ cells and the proportion of NKG2A⁺ or CD16⁺ cells (Fig. S3).

3.3. Granule mobilization only correlates with storage of perforin in cord blood Vδ2 cells

Following TCR-mediated restimulation (16 days after the first stimulation), the expression of CD107a, a marker of cytotoxic granule mobilization, showed expression patterns similar to perforin, with an enrichment of CD107a⁺ cells in the CD56⁺ and in the PD-1⁺ subset of Vδ2 cells (*p* < 0.0001, data not shown). As for perforin, the highest

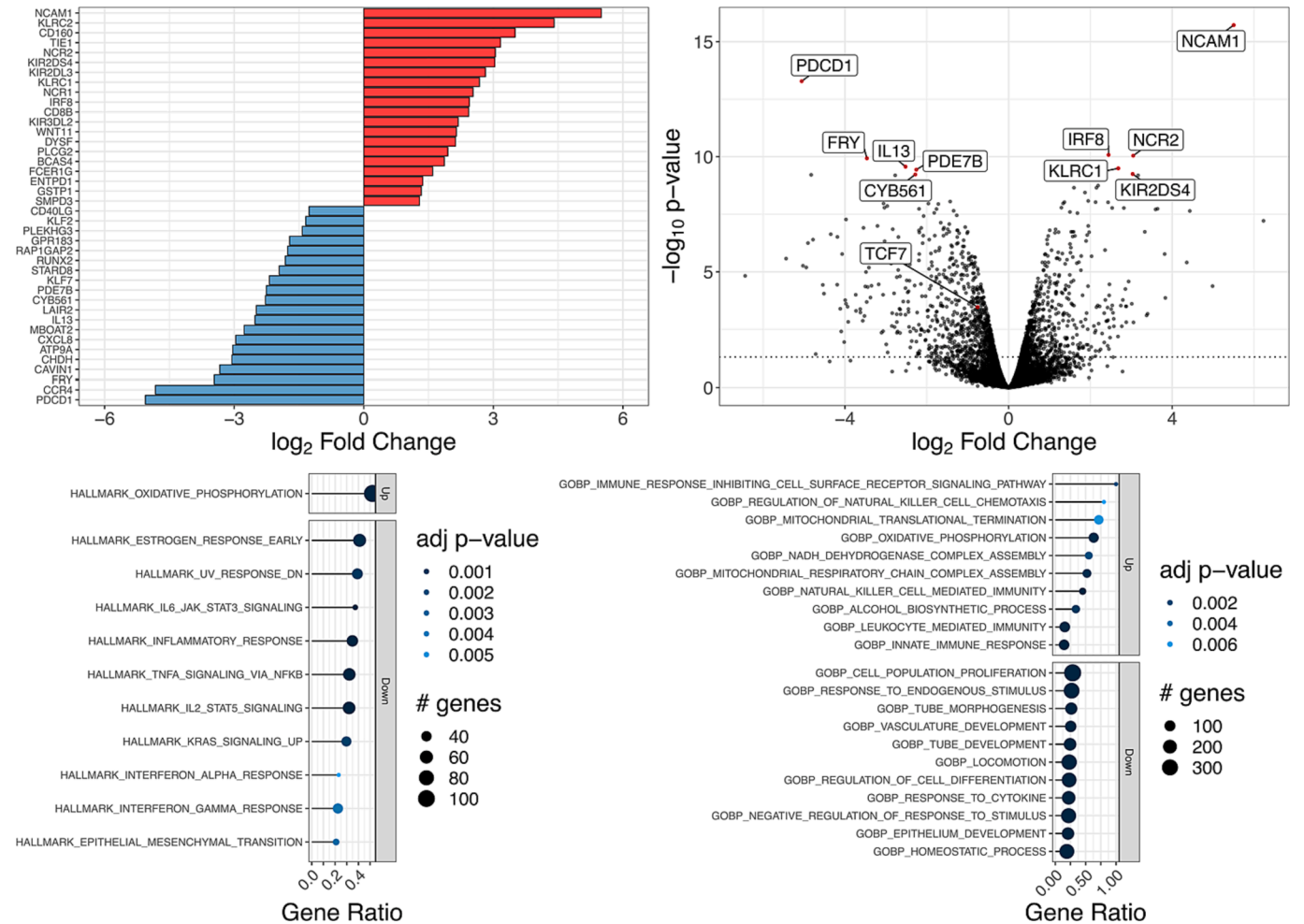


Fig. 6. PD-1⁺CD56⁺ cells display transcriptional differences compared to PD-1⁺CD56⁻ cells. Fourteen days after stimulation, PD-1⁺CD56⁻ (left) or PD-1⁺CD56⁺ (right) cells were sorted on a FACS Aria II. Total RNA was extracted and shipped to Novogene for bulk RNA sequencing. Differential expression and gene set enrichment analyses were performed in R using limma, fgsea and collections from the Molecular Signature Database. A) Top ranked genes (for adjusted p-value) differentially upregulated (red) and downregulated (blue) genes in PD-1⁺CD56⁺ as compared to PD-1⁺CD56⁻ cells. The y-axis lists the gene symbols whereas the x-axis lists the magnitude of change in the expression. B) Volcano plot of differential expression (DE) analysis. Ten of the top ranked genes were plotted in red and labeled. The y-axis indicates the adjusted p-value of the DE analysis whereas the x-axis the magnitude of change in the expression of PD-1⁺CD56⁺ as compared to PD-1⁺CD56⁻ cells. C & D) Top ranked (for adjusted p-value) enriched pathways for the Hallmark (C) and The Gene Ontology Biological Processes Gene Sets (D). The upper and bottom panels list up- and down-regulated pathways, respectively. The x-axis indicates the ratio of leading-edge genes over the total genes in the pathway; p-values and total number of genes of the pathway are mapped to size and color of the symbols, respectively.

Table 2
Summary of IPA findings comparing PD-1⁺CD56⁻ and PD-1⁻CD56⁺ cell transcriptomes.

Top Canonical pathways		
Name	p-value	Overlap
Cardiac Hypertrophy Signaling (Enhanced)	3.40E-20	17.9 % (97/542)
Natural Killer Cell Signaling	8.42E-18	25.6 % (51/199)
Th1 and Th2 Activation Pathway	1.03E-17	27.3 % (47/172)
Erythropoietin Signaling Pathway	5.01E-15	24.9 % (44/177)
HMGB1 Signaling	1.35E-14	25.1 % (42/167)
Molecular and Cellular Functions		
Name	p-value range	# Molecules
Cellular Movement	2.93E-14—6.33E-60	539
Cellular Development	9.33E-15—1.60E-50	540
Cellular Growth and Proliferation	1.93E-14—1.60E-50	510
Cellular Function and Maintenance	3.76E-14—5.48E-45	609
Cell-To-Cell Signaling and Interaction	2.62E-14—2.67E-40	378
Physiological System Development and Function		
Name	p-value range	# Molecules
Hematological System Development and Function	2.68E-14—1.46E-51	511
Tissue Morphology	2.68E-14—1.46E-51	428
Lymphoid Tissue Structure and Development	2.68E-14—1.60E-50	393
Immune Cell Trafficking	2.48E-14—2.04E-43	331
Hematopoiesis	9.33E-15—4.76E-41	262
Molecules		Expr. Value
NCAM1		1.90E-16
PDCD1		5.22E-14
IRF8		8.29E-11
NCR2		8.99E-11
Summary of Analysis - de_genes		Expr. Value
FRY		1.17E-10
IL13		2.70E-10
KLRC1		3.18E-10
PDE7B		3.65E-10
KIR2DS2 (includes others)		5.61E-10
CYB561		5.91E-10

frequencies of CD107a⁺ cells were found in the PD-1⁻CD56⁺ subset, while the lowest were found in the PD-1⁺CD56⁻ subset (Fig. 4 A-B). The proportion of CD107a⁺ cells displayed a moderate negative correlation with the proportion of total PD-1⁺ and PD-1⁺CD56⁻ cells ($p = 0.03$ and $p < 0.0001$, respectively, Fig. 4 C and D), a weak correlation with the proportion of CD56⁺ cells ($R = 0.29$ with $p = 0.024$, data not shown) and a moderate positive correlation with the PD-1⁻CD56⁺ subset ($p < 0.0055$, Fig. 4 D). CD107a was the only major variable in the study to show significantly different distributions based on sex of the infants. The proportion CD107a⁺ Vδ2 cell was, in fact, mildly but significantly higher for female neonates after BCG-mediated expansion compared to male neonates after the same treatment (Fig. S4). No differences between male and females were observed for specimens after ZOL stimulation and for the pooled CD107a values (not shown).

Since the percentages of CD107a⁺ and perforin⁺ Vδ2 cells were surprisingly similar in most specimens after expansion (Fig. 5 A), we tested if an association exists between these two parameters using a linear regression model. We observed a moderate positive correlation at birth, but not in 12-month-old infants ($p < 0.0001$ and not significant, respectively, Fig. 5 B). We thus hypothesized that the apparent correlation between CD107a and the phenotypic subsets PD-1⁻CD56⁺ and

PD-1⁺CD56⁻ is due to the fact that the proportion of perforin⁺ cells independently correlates with all three variables. To test this, we performed the adjusted analyses listed in Table 1. Even in adjusted analyses including CD107a as a variable, perforin was independently associated with PD-1⁺CD56⁻ and PD-1⁻CD56⁺ (Table 1: Models 1 and 2). Specimens with higher proportions of perforin⁺ Vδ2 cells were more likely to have lower PD-1⁺CD56⁻ levels ($p = 0.002$) and higher PD-1⁻CD56⁺ levels ($p = 0.02$). The proportion of perforin⁺ cells was predicted to increase by 0.28 with every unit increase in the proportion of PD-1⁻CD56⁺ cells and decrease by 0.4 with every unit increase in the proportion of PD-1⁺CD56⁻ cells. Conversely, CD107a in the adjusted analyses was only associated with Perforin, not with PD-1⁺CD56⁻ or PD-1⁻CD56⁺ (Table 1: Models 3 and 4). Lastly, we ran models where the outcomes were PD-1⁺CD56⁻ or PD-1⁻CD56⁺ (Table 1: Models 5 and 6), observing results consistent with Models 1 and 2.

3.4. PD-1⁻CD56⁺ cells display several differentially regulated genes compared to PD-1⁺CD56⁻ cells

Since the PD-1⁻CD56⁺ and PD-1⁺CD56⁻ subsets have significantly different cytotoxic potential, we sought to analyze their transcriptional profile to test whether a gene signature associated with stronger cytotoxic potential or different regulatory pathways would be evident in the PD-1⁻CD56⁺ subset. The cells sorted in FBS for quality control purposes revealed purities consistently above 95 % (Fig. S5). RNA-seq analysis showed that several genes encoding NK receptors (such as KLRC2, NCR2, KIR2DS4, NCR1) were overexpressed in the PD-1⁻CD56⁺ population (Fig. 6 A and B). Pathway analysis confirmed that genes related to NK-mediated immunity were over-expressed in the CD56 single positive subset. Interestingly, one of the most upregulated pathways was oxidative phosphorylation, suggesting that metabolism may be linked to cytotoxic effector differentiation in neonatal Vδ2 cells. Conversely, genes involved in the response to cytokines and proliferation were expressed at higher levels in the PD-1⁺CD56⁻ subset (Fig. 6 C and D). This subset also displayed higher expression levels of IL-13 and CCR4, which are associated with Th2 function [61–64], and the transcription factor Tcf7 (although the difference in Tcf7 expression is small relative to the other genes) (Fig. 6).

We then performed the IPA to identify biological pathways enriched in the differentially expressed genes (Table 2). The top differentially regulated canonical pathways included Natural Killer Cell Signaling; Th1 and Th2 Activation Pathway; Erythropoietin Signaling Pathway, and HMGB1 Signaling. Overall, the transcriptomic profiles of CD56 single positive and PD1 single positive cells were consistent with a functional difference between these subsets, with an overexpression of transcripts related to NK function in the former.

3.5. IRF8 is expressed at higher levels in PD-1⁻CD56⁺ cells than in PD-1⁺CD56⁻ cells

Among the most differentially regulated genes, the transcription factor IRF8 was overexpressed in the PD-1⁻CD56⁺ subset (Fig. 6 A-B, Table 2). This gene has been shown to drive CD8 T cell effector differentiation [65] and NK cell maturation [66], and thus might be involved in the differentiation of cytotoxic Vδ2 cells. We employed RT-PCR to confirm IRF8 differential expression, sorting the Vδ2 cell subsets 10 and 14 days after stimulation (D10 and D14). As expected, IRF8 mRNA levels were higher in the PD-1⁻CD56⁺ subset at both time points but showed a more significant difference on D10 compared to D14. Additionally, IRF8 transcripts in the PD-1⁻CD56⁺ subset decreased significantly from D10 to D14 (Fig. 7 A, $p = 0.028$). Interestingly, adult Vδ2 cell subsets on D10 displayed comparable IRF8 expression, significantly lower than the neonatal PD-1⁻CD56⁺ fraction ($p < 0.0001$ Fig. 7 A). Zbtb32, a master regulator of virus-driven NK cell proliferation, is directly controlled by IRF8 in NK cells [67]. We thus analyzed Zbtb32 mRNA levels in RT-PCR to test whether its expression pattern mirrors IRF8. Zbtb32 mRNA was

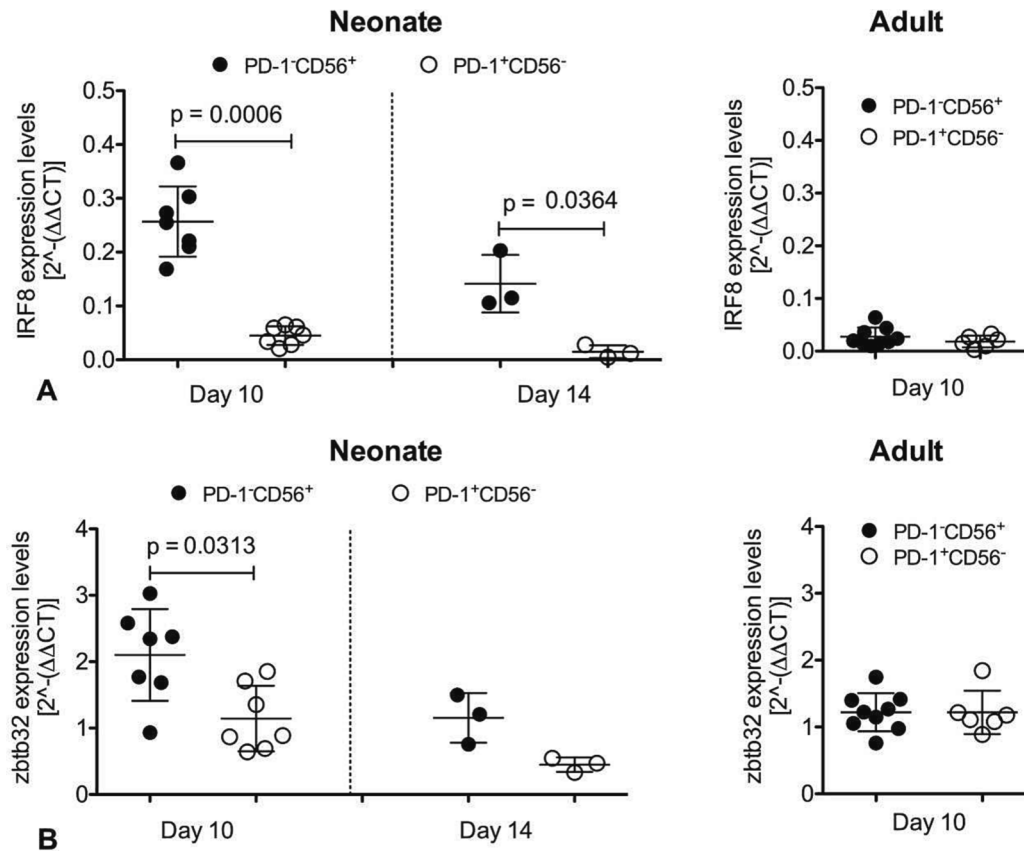


Fig. 7. PD-1⁺CD56⁺ cells display higher levels of IRF8 and Zbtb32 mRNA than PD-1⁺CD56⁻ cells. Ten or fourteen days after stimulation, PD-1⁺CD56⁻ (left) or PD-1⁺CD56⁺ (right) cells were sorted and RNA was extracted for real time PCR analysis. mRNA levels for sorted Vδ2 cell subsets were evaluated using the comparative method. The scatter plots display the relative expression of A) IRF8 and B) Zbtb32 in neonatal (left) and adult (right) Vδ2 cell subsets 10 or 14 days after stimulation, showing means, SD and individual values for each variable at both time points.

present at significantly higher levels in the PD-1⁺CD56⁺ fraction on D10. However, the difference in Zbtb32 expression between the two Vδ2 cell subsets was not as marked as it was for IRF8 and did not reach statistical significance on D14 (Fig. 7 B). No difference between subsets was observed for adult Vδ2 cells (Fig. 7 B).

In order to validate the observation on IRF8 differential expression, we stained a subset of CBMC specimens with anti-IRF8 antibody, acquiring the samples on an Aurora Cytex analyzer for increased sensitivity. The flow cytometry analyses clearly showed that cord blood Vδ2 cells express IRF8 7 and 10 days after stimulation, with a significant decrease in the proportion of positive cells between day 7 and 10 (Fig. 8). Importantly, the PD-1⁺CD56⁺ subset displayed at both time points the highest proportion of IRF8⁺ cells, while the PD-1⁺CD56⁻ subset displayed the lowest (Fig. 8). The expression of IRF8 protein across Vδ2 cell subsets followed the same pattern as Perforin, though the kinetics were different.

4. Discussion

Following up on our observations on PD1 regulatory function in cord blood Vδ2 cells [38,42], in the current study we aimed to assess neonatal Vδ2 cell cytotoxic potential in relation to PD-1 expression. This study shows that the concomitant assessment of PD-1 and CD56 expression identifies subsets of cord blood Vδ2 cells with different cytotoxic potential, in particular the PD-1⁺CD56⁺ subset with high potential and the PD-1⁺CD56⁻ with low potential. Transcriptomic analysis highlighted that several genes related to NK-mediated cytotoxicity are expressed at higher levels in the PD-1⁺CD56⁺ subset compared to the PD-1⁺CD56⁻. In addition, the transcription factor IRF8 is overexpressed in the CD56

single positive cells up to 14 days after stimulation and we speculate that it may play a role in the differentiation of cytotoxic Vδ2 effectors.

A PCA to evaluate the composite effect of the distribution of Vδ2 across the four CD56/PD-1 subsets on perforin expression suggested that specimens with the lowest perforin levels tend to cluster separately from the specimens with intermediate or high perforin levels. Considering directionality and absolute values of the eigen vectors for PC1 and PC2, most of the specimens in the low perforin stratum have both high frequencies of PD-1⁺ (PD-1⁺CD56⁻) and low frequencies of CD56⁺ (PD-1⁺CD56⁺) cells. The percentage of PD-1⁺CD56⁻ cells explains the largest fraction of variance in perforin distribution among all the variables included in our linear regression models. With multivariable linear regression models, we were able to clarify that the proportion of CD107a⁺ cells after restimulation significantly correlates with the proportion of perforin⁺ cells, but not with the proportion of PD-1⁺CD56⁻ or PD-1⁺CD56⁺ cells. This association is only evident for neonatal, not infant or adult cells. Since perforin is not essential for degranulation to occur [68], the presence of this mediator may indicate that the granules have completed their biogenesis and are ready to be released, while absence of perforin may identify granules that are not completely mature, hence not ready to be released.

Analysis of perforin expression in relation to other NK cell markers confirmed that both CD16 and NKG2A are good predictors of cytotoxic potential at the single cell level, since they identify subsets highly enriched in perforin⁺ cells, but are not good predictors of the cytotoxic potential of cord blood Vδ2 cells as a population. In fact, the proportion of perforin⁺ cells among Vδ2 lymphocytes does not correlate with the proportions of NKG2A⁺ or CD16⁺ cells, which in most cases are substantially lower.

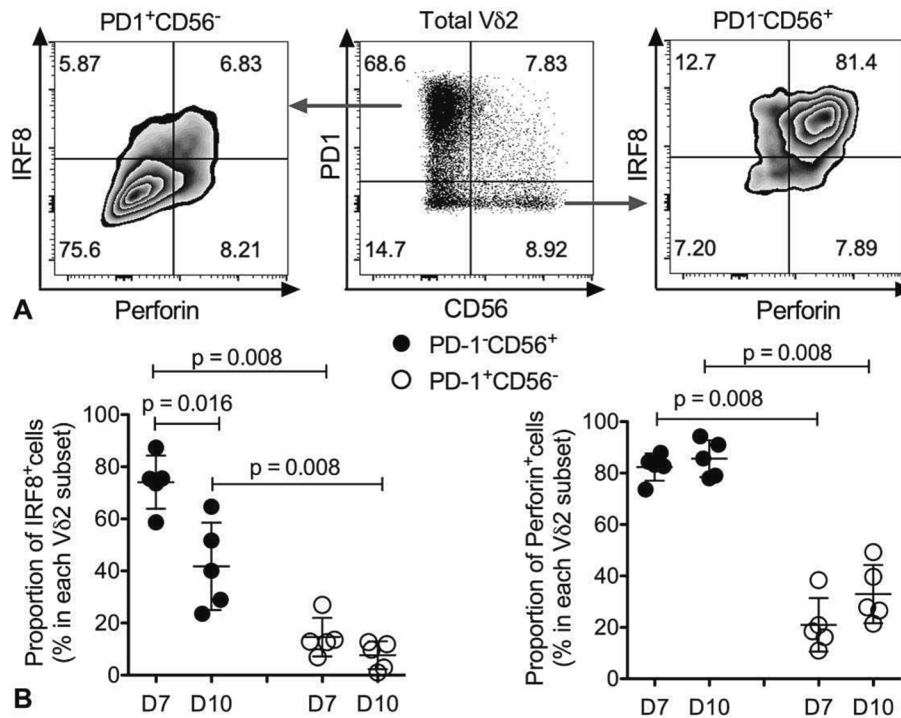


Fig. 8. The proportion of IRF8⁺ Vδ2 cells is higher in the PD-1⁻CD56⁺ subset than in the PD-1⁺CD56⁻ subset. Seven and ten days after ZOL stimulation, CBMC were stained with anti-IRF8. A) The zebra plots display the expression of Perforin and IRF8 in the PD-1⁺CD56⁻ (left) or PD-1⁻CD56⁺ (right) subset of total Vδ2 cells (central dot plot). B) The scatter plots summarize the results of the experiment showing mean, SD and individual values for the proportion of IRF8⁺ (left) or perforin⁺ (right) Vδ2 cells at 7 and 10 days after ZOL stimulation.

A significant limitation of this study is that we relied on cells expanded *in vitro*. However, the enrichment of perforin⁺ cells among total PD-1⁻ and PD-1⁻CD56⁺ Vδ2 lymphocytes was also observed *ex vivo*, for the few specimens that had enough perforin⁺ cells to analyze their distribution across the subsets identified by PD1 and CD56. Additionally, the protocol we used for expansion has been standardized and used for previous studies [38,42]. The presence of a large proportion of PD-1⁺ resting Vδ2 cells and the negative association between PD-1 and cytotoxic potential are observed only for neonatal Vδ2 lymphocytes, not for their counterparts in 12-month-old infants or adults, pointing at unique properties of these cells in cord blood.

Among all the major variables in the study only the proportion of CD107a⁺ Vδ2 cells after stimulation with BCG showed a small but significant difference between male and female infants. The proportion of perforin⁺ Vδ2 cells after the same treatment showed a similar trend but the difference did not reach statistical significance. These observations do not conclusively rule out differences between sexes for key Vδ2 cell variables at birth, but suggest that the differences, if they exist, are small and not appreciable in this study with a relatively small sample size.

A limitation of this study is that we cannot conclude whether PD-1 engagement plays a role in the differentiation of cell subsets with different cytotoxic potential, or whether this inhibitory receptor simply serves as a marker. Multiple lines of evidence also suggest that PD-1 engagement does not occur at the time we assess the cells for perforin expression (16 days after stimulation). First, no cells stain positive for PD-L1 and few monocytes/macrophages are left in the culture (data not shown). Most importantly, we previously showed that engagement of PD-1 during TCR stimulation effectively inhibits cytokine production to an extent directly proportional to the proportion of PD-1⁺ cells [42]. In presence of PD-1 engagement we would thus expect the PD-1⁺ subset to contain a significantly lower percentage of cytokine-producing cells, which is clearly not the case. Overall, our data suggests that PD-1 expression on resting Vδ2 cells may represent a marker of a unique differentiation stage at birth. We speculate that PD-1⁺CD56⁻ cord blood

Vδ2 cells, which appear to express the *Tcf7* transcript at higher levels than PD-1⁻CD56⁺ cells, may share some properties of stem-like PD-1⁺ CD8 T cells [69,70]. However, additional experiments are needed to exclude a direct role of PD-1 in the differentiation of Vδ2 cells with low cytotoxic potential.

The results of bulk RNA sequencing confirm that PD-1⁻CD56⁺ cells overexpress genes related to NK cell biology and responses, including several NK receptors. Interestingly, PD-1⁻CD56⁺ cells may be less responsive to homeostatic cytokines and less prone to proliferation. This suggests that they could represent a subset further along in the differentiation toward effector memory cells compared the PD-1⁺CD56⁻ subset. Functional assays performed with purified cell subsets are needed to clarify this issue.

IRF8 overexpression at the transcript and protein level in the most cytotoxic subset (as late as 14 and 10 days after stimulation, respectively) is intriguing, as it suggests that this transcriptional regulator may be involved in the establishment of a cytotoxic effector program for Vδ2 cells in early life. Since IRF8 primarily acts as a negative regulator, it may help repress the default transcriptional program of fetal Vδ2 cells, which possibly regulates the acquisition of cytotoxic potential. Chromatin accessibility (ATAC-seq) and promoter occupancy studies (ChIP-seq) will be needed to determine whether IRF8 indeed helps program neonatal Vδ2 cells into cytotoxic effectors. The observation that IRF8 levels are lower in adult compared to neonatal Vδ2 lymphocytes is most likely due differences in expression kinetics, with faster down-modulation of its expression in adult cells than in cord blood Vδ2 cells (similar to what we previously observed for PD-1 expression [42]). Time course analyses of IRF8 mRNA levels, which would help address this issue, will be challenging due to the relatively low proportion of neonatal PD-1⁺ Vδ2 cells and their propensity to undergo activation induced cell death if sorted shortly after stimulation [42]. In an effort to maximize RNA yields, we sorted the subsets of interest 14 days after stimulation, when Vδ2 cells have entered the resting phase, and collected the fractions in trizol to prevent loss of RNA due to

fragmentation in case of cell death. Experiments requiring large numbers of viable cells or cells sorted earlier after stimulation will certainly pose a technical challenge. CITE-seq analyses of Vδ2 cells enriched via magnetic negative isolation may provide the most effective and informative approach for future experiments.

Our PD-1-related observations only apply to cord blood Vδ2 cells. In infants and adults, the proportion of PD-1⁺CD56⁺ cells among resting Vδ2 cells is low (often less than 5 %) and the fraction of perforin⁺ cells is similar in the PD-1⁺ and PD-1[−] subset [38]. Overall, our results suggest that PD-1 expression is negatively associated with the development of Vδ2 cell cytotoxic potential in the perinatal period. In the first few weeks of life, the acquisition of NKG2A expression [39], possibly resulting from exposure to higher levels of microbial PAg, may coincide with a shift in the mechanisms regulating Vδ2 cell cytotoxicity as an adaptation to a different ontogenetic stage.

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CRediT authorship contribution statement

Haoting Hsu: Investigation, Validation, Visualization, Writing – original draft. **Claudio Zanettini:** Formal analysis, Writing – review & editing, Visualization, Data curation. **Modupe Coker:** Formal analysis, Visualization, Writing – review & editing, Data curation. **Sarah Boudova:** Investigation, Resources, Writing – review & editing. **David Rach:** Investigation, Writing – review & editing. **Godfrey Mvula:** Investigation, Resources, Writing – review & editing. **Titus H. Divala:** Resources, Supervision, Writing – review & editing. **Randy G. Mungwira:** Resources, Supervision, Writing – review & editing. **Francesca Boldrin:** Resources, Writing – review & editing. **Giulia Degiacomi:** Resources, Writing – review & editing. **Laura Cioetto Mazzabò:** Resources, Writing – review & editing. **Riccardo Manganelli:** Resources, Writing – review & editing. **Miriam K. Laufer:** Funding acquisition, Supervision, Writing – review & editing, Resources. **Yuji Zhang:** Formal analysis, Validation, Writing – review & editing. **Luigi Marchionni:** Formal analysis, Writing – review & editing. **Cristiana Cairo:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Writing – original draft, Writing – review & editing.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cellimm.2023.104797>.

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